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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 11** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Case Study Question 12(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study question of 5 marks with internal choice.
7. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. From the top of a 20 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its base is 30°. The horizontal distance between the building and the tower is: [1]

(A) $20\sqrt{3}$ m (B) $\frac{20}{\sqrt{3}}$ m (C) 40 m (D) 60 m

Q2. In Q1, the total vertical height of the tower is: [1]

(A) 60 m (B) 80 m (C) $20\sqrt{3}$ m (D) 100 m

Q3. An electric pole is 10 m high. A steel wire tied to the top of the pole is affixed to the ground at an angle of 45°. The length of the wire is: [1]

(A) $10\sqrt{2}$ m (B) $10\sqrt{3}$ m (C) 20 m (D) $5\sqrt{2}$ m

Q4. A steel wire of length 20 m is tied from the top of an electric pole to the ground making an angle of 60°. The height of the pole is: [1]

(A) $10\sqrt{3}$ m (B) 10 m (C) $20\sqrt{3}$ m (D) $\frac{20}{\sqrt{3}}$ m

Q5. **Assertion (A):** As an observer walks towards the foot of a vertical monument, the angle of elevation of the top of the monument continuously decreases.
Reason (R): As distance d decreases, $\tan\theta = \frac{h}{d}$ increases, which implies θ increases for acute angles. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. If the angle of elevation of the top of a tower from a distance of 100 m from its foot is 30°, find the height of the tower. [2]

Q7. A vertical flagstaff of height 5 m is fixed on top of a tower. From a point on the ground, the angles of elevation of the top and bottom of the flagstaff are 60° and 45° respectively. Find the height of the tower. [Take $\sqrt{3} = 1.732$] [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. Two lamp posts of equal heights are situated on opposite sides of a road 60 m wide. From a point on the road between them, the angles of elevation of the tops of the posts are 60° and 30°. Find the height of the posts and the position of the observation point. [3]

Q9. A surveillance helicopter hovers stationary at an altitude of 600 m. The angle of depression of a stationary patrol vessel is 30°. If the helicopter flies horizontally towards the vessel at a speed of 180 km/h, find the angle of depression after 10 seconds. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

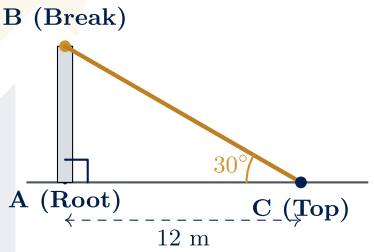
Q10. An observer on top of a cliff 150 m high observes two boats in the sea moving directly towards the foot of the cliff along the same radial line. If the angles of depression of the boats change from 30° to 45° and 45° to 60° respectively, find the distances travelled by both boats. [Take $\sqrt{3} = 1.732$] [5]

Q11. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h. At a point on the plane, the angles of elevation of the bottom and top of the flagstaff are α and β respectively. Prove that the height of the tower is: [5]

$$\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$$

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Cyclonic Storm and Coconut Palm Fracture



During a severe tropical cyclone along the coastal belt of Odisha, a tall vertical coconut palm tree was fractured by gale-force winds. The stem snapped at a certain height above the ground (B) without completely detaching from the trunk base (A).
The crown top of the palm tree (C) struck the flat ground at a horizontal distance of 12 m from the root (A), making an angle of inclination of 30° with the ground level.

Based on the above mathematical scenario, answer the following questions:

(i) Find the vertical height above the ground at which the palm tree snapped (AB). [Take $\sqrt{3} = 1.732$] [1]

(ii) Find the length of the broken trunk section that bent over to touch the ground (BC). [2]

(iii) Calculate the total original standing height of the coconut palm tree before the cyclonic storm. [2]

— OR —

If the tree had broken such that its top struck the ground at an angle of 45° at a distance of 8 m from its root, find its original standing height. [Take $\sqrt{2} = 1.414$] [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★